Allen Choi

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EDUCATION

Doctor of Philosophy (Ph.D.) in Statistics 2026 – 2031
University of California, Santa Cruz *Santa Cruz, CA*

Master of Science (M.S.) in Statistics, with Distinction June 2026
California Polytechnic State University *San Luis Obispo, CA*

- **GPA**: 3.84/4.00

Bachelor of Science (B.S.) in Statistics, Minor in Data Science, Magna Cum Laude June 2026
California Polytechnic State University *San Luis Obispo, CA*

SKILLS

Languages: R, Python, SQL, Java, SAS
Libraries: pandas, PyTorch, TensorFlow, scikit-learn, DuckDB, PySpark
Software: Tableau, GitHub, Colab

EXPERIENCE

Graduate Teaching Assistant Sept 2026 – Present
University of California, Santa Cruz *Santa Cruz, CA*

- Lead **3 discussion sections per week** for STAT 17, facilitating group work and reinforcing lecture material
- Hold **office hours** and **exam review sessions**; help with **exam grading and proctoring**

Data Analyst Intern June 2025 – Aug 2025
GALLO *Modesto, CA*

- Developed and deployed a comprehensive data quality monitoring tool utilizing **Python**, **SQL**, and **statistical analysis**, streamlining the resolution of data quality issues for GALLO's data governance team
- Automated data quality insights by integrating a **Large Language Model** to analyze monitoring results and present findings concisely, improving the team's ability to quickly understand/act on data quality issues
- Ensured successful tool adoption through regular stakeholder communication, **Git**-based version control, and hands-on training delivery within an **Agile/Scrum** development framework

Undergraduate Data Science Researcher June 2024 – Aug 2024
University of California, Irvine *Irvine, CA*

- Analyzed large-scale **spatial transcriptomics** data in **R** to study how vesicant skin injury progresses over time, as part of a three-student research team
- **Presented findings** at the closing symposium of **ISI-BUDS**, an NIH-funded Summer Institute in Biostatistics (SIBS) program led by UCI Statistics faculty

Statistics Learning Tutor Oct 2023 – June 2026
California Polytechnic State University, San Luis Obispo *San Luis Obispo, CA*

- Facilitated tutoring in Cal Poly's Learning Support Center, supporting up to **10 different statistics courses**
- Provided **individualized tutoring** across **nearly 500 sessions** in statistics fundamentals and software, helping students significantly improve understanding and boost performance in statistics courses

PROJECTS

Facial Recognition Model (Python - TensorFlow/PyTorch)

- Investigated **bias in AI predictions** by training and comparing **Multi-Layer Perceptron (MLP)**, **Convolutional Neural Network (CNN)**, and **ResNet-50** models in TensorFlow/PyTorch for ethnicity and gender classification from face images
- Addressed prediction bias through **iterative model evaluation and refinement**, notably improving accuracy

Cancer Prediction Model (Python - scikit-learn)

- Cleaned and preprocessed census data of **400k+** total participants' demographic information
- Utilized scikit-learn to construct and fine-tune a K-Nearest Neighbors, Random Forest, and an ensemble model to classify cancer history in people, optimizing hyperparameters to maximize recall scores